MOHAMMAD RABBY

AI Engineer | Data Engineer

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LinkedIn | GitHub | LeetCode | HackerRank

PROFESSIONAL SUMMARY

AI and Data Engineer with hands-on experience turning real business problems into production-ready systems. I build LLM-powered workflows using the Claude and OpenAI APIs, automate data pipelines with Python and SQL, and ship reliable end-to-end solutions backed by proper testing. I work comfortably across the full stack from ETL and dashboards to browser automation and AI agents and I take AI-generated code seriously enough to review, debug, and validate it before it ships.

TECHNICAL SKILLS

- **Languages**: Python, SQL, R, PySpark, JavaScript (basic), C, HTML, CSS
- **AI & LLM Engineering**: Claude API, OpenAI API, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Application Development, AI Agents, AI Workflow Automation
- **AI-Assisted Development**: Claude Code, GitHub Copilot, AI-Assisted Test Generation, AI-Driven Root-Cause Analysis
- **QA & Test Automation**: Test Case Design, Regression & Functional Testing, Defect Tracking, Playwright, Selenium, Pytest, REST API Testing, API Integration
- **Machine Learning & Data Science**: Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, PyTorch, LSTM, ARIMA, Matplotlib, Seaborn
- **Data Engineering & Databases**: ETL Pipelines, Apache Airflow, Docker, MySQL, PostgreSQL, Web Scraping, asyncio, JSON
- **BI, Tools & Platforms**: Power BI, Tableau, Excel, Git/GitHub, VS Code, PyCharm, Jupyter Notebook, Agile, CI/CD, Azure, Linux

EXPERIENCE

Data Analyst **Oct 2025 – Present**
Data Insight LLC. *Dhaka, Bangladesh*

- Build and automate dashboards, reports, and data workflows in Python, SQL, Excel, and Power BI; manage end-to-end AI and automation projects; validate large datasets for accuracy; and collaborate cross-functionally to deliver decision-ready insights.

Data Analyst Intern **Mar 2025 – Aug 2025**
CodeAlpha Technologies *Lucknow, India (Remote)*

- Wrote and optimized complex SQL queries for efficient data extraction, integration, and validation across large relational datasets.
- Cleaned, validated, and analyzed large datasets to generate actionable insights that supported business decisions.
- Automated recurring reporting tasks and built interactive Power BI and Excel dashboards, reducing manual report-preparation time.

PROJECTS

AI-Powered B2B Lead Generation Platform | Live | *Python, FastAPI, OpenAI API, PostgreSQL, Next.js, Playwright* **2026**

- Built LLM website-analysis pipeline extracting company offerings for auto-filled onboarding, with heuristic fallback and input capping cutting token spend.
- Developed AI Campaign Builder generating cold-outreach sequences as drafts only, with validator blocking unfilled placeholders from reaching real recipients.
- Shipped natural-language audience builder converting free text into structured JSON filters; validated with 87 test suites and mutation testing.

Edge-AI Blind Assist System with Object Detection | GitHub | *Python, YOLO, OpenCV, EasyOCR, Jetson Nano* **Feb 2026**

- Engineered a YOLO and OpenCV edge-AI computer vision system delivering real-time object detection with audio feedback for visually impaired users.
- Integrated EasyOCR and bi-directional voice APIs to deliver interactive, multilingual audio feedback for blind users.
- Deployed an optimized, GPU-accelerated Python pipeline onto a Jetson Nano for real-time on-device detection.

Real-Time Sports News Intelligence System | GitHub | *Python, asyncio, Playwright, Claude API, SQLite, Telegram* **2026**

- Engineered an asynchronous scraping pipeline (aiohttp, 50 concurrent workers) monitoring 500+ news sources across 48 countries 24/7, with Playwright browser automation and rotating proxies for JS-heavy and bot-protected sites.
- Built a multilingual NLP detection engine (10+ languages, stemming and fuzzy matching) feeding a weighted 0–10 risk-scoring algorithm with an ML confidence filter, delivering color-coded Telegram alerts matched to live fixture data via REST APIs.
- Integrated the Claude API with structured prompt engineering to convert raw multilingual scraped content into concise, human-readable English alert summaries.
- Ensured production reliability with SQLite-backed deduplication, automatic failing-source recovery, and a 204-case pytest suite (100% passing, pyflakes-clean) covering scraping, scoring, and alerting logic.

AI Voice Booking Agent for Courier Services | GitHub | *Python, FastAPI, Vapi, Twilio, Google Maps API, Render* **2026**

- Built a 24/7 AI phone booking agent using FastAPI and Vapi (GPT-4.1), quoting courier jobs in real time from Google Maps mileage and tiered pricing.
- Automated end-to-end booking through async webhooks, concurrently creating Cal.com calendar entries, Google Sheets records, Twilio SMS and Resend email confirmations for customers and the office.
- Deployed to Render with fault-tolerant fallbacks, automated failure alerts and a 129-case pytest suite, handling live customer calls and bookings for a UK courier company.

Stock Market Forecasting with Time-Series Financial Data | GitHub | *Python, Scikit-learn, LSTM, ARIMA* **Sep 2024**

- Developed a forecasting pipeline in Python using LSTM (deep learning) and ARIMA (statistical) models to predict stock price movements.
- Engineered features from time-series candlestick data and validated models using RMSE/MAE performance metrics.
- Conducted a comparative analysis of deep learning vs. statistical forecasting models to support investment decision-making.

E-Commerce Customer Order & Revenue Dashboard | GitHub | *Python, SQL, Power BI, Excel, Pandas* **Jun 2023**

- Designed an ETL pipeline in Python and SQL to clean, aggregate, and analyze customer transactions and order history.
- Built interactive Power BI dashboards covering RFM segmentation, sales-performance KPIs, and customer retention.
- Identified high-value customers, revenue leakage, and repeat-purchase trends, enabling actionable business insights.

EDUCATION

Khulna University **Jan 2020 – Feb 2025**
B.Sc.(Engg) in Electronics & Communication Engineering (ECE) — GPA: 3.28 *Khulna, Bangladesh*

- **Relevant Coursework**: Python, Java, C, Machine Learning, Artificial Intelligence, Data Structures

CERTIFICATIONS

- **Machine Learning Specialization** — IBM 2024
- **Complete Data Analyst Bootcamp** 2024
- **Databases & SQL for Data Science with Python** 2025
- **Agile Project Management** 2025

ACHIEVEMENTS & PROBLEM SOLVING

- Awarded **Best Employee of the Data Team** 3 times within 8 months at Betopia Group, in recognition of consistent, high-impact delivery.
- Regularly solve data structures, algorithms, and SQL challenges on LeetCode and HackerRank to sharpen problem-solving skills.
- Maintain a public portfolio of end-to-end automation, AI, and analytics projects with documented code and test suites on GitHub.
- **Languages**: English (Professional Working Proficiency), Bengali (Native)